

TugOS · Vertical Architecture Blueprint

OSL Orchestrator Model — Applied to Maritime Operations

OPEN SCAFFOLD LABS · VERTICAL APPLICATION BLUEPRINT · Version 2.0 · June 2026 · App 24 · Table Prefix: tug_

TugOS is App 24 in the OSL ecosystem. Every architectural primitive — the three-surface model, the MCP layer, the Catalog of Authority, the genetic template — maps onto tugboat operations. Version 2 makes one deliberate deviation from the template (database isolation) and corrects the regulatory and licensing assumptions of v1.

v2 architecture decisions. (1) TugOS runs in a **dedicated Supabase project**, not the shared 23-app instance — SOC 2 and port-authority security reviews require a clean tenant isolation story, enforced with per-tenant RLS. (2) The compliance core is **46 CFR Subchapter M / TSMS**. (3) Real-time AIS comes from **commercial APIs**; the CC BY-NC MarineTraffic toolbox is excluded from the product. (4) Mobile auth uses refresh-token grace so offline-first sessions survive 15-minute web tokens. (5) A human **maritime domain advisor** complements the Catalog of Authority.

01 · The Genetic Template — Deploy in Under 5 Minutes

FRAMEWORK AS A SERVICE — shared across all 23 OSL apps: @openscaffold/core, @openscaffold/integrations, scaffold.config.js. Run deploy-new-app.sh tug_ and TugOS inherits StandardHeader, the Scaffold Beacon AI widget, role-based sidebar, JWT auth, theme system, CRUD generation standards, and all integration adapters (pdf-lib, PptxGenJS, React Email, DocuSeal, LlamaIndex). The one v2 deviation: the Supabase connection points at a dedicated TugOS project (see 07).

Domain Layer — TugOS Domain UI

Dispatch · Crew · Fuel · Maintenance · Billing — vessel dispatch board, crew hitch scheduler, per-voyage fuel log, Sub M/TSMS compliance tracker, invoice generator.

Expertise Layer — Catalog of Authority + Domain Advisor

Validated MCP connections to the authorities a maritime expert consults (see 06), **plus** a retained human advisor — a retired port captain or ops manager — for workflow validation, sales credibility, and conference presence. Data connections answer regulatory questions; the advisor answers 'would a captain actually use this?'

Intelligence Layer — Paperclip + Claude API

OSL's production AI orchestration. Fuel anomaly detection · maintenance prediction · crew cert risk scoring · auto-invoice from job log · dispatch optimization.

02 · The Three-Surface Architecture

- **Surface 1 — UI (human-facing):** React 19 + Vite + Tailwind. Role-filtered sidebar: Dispatcher sees the job board, Captain sees the vessel log, Fleet Manager sees P&L. Roles: Dispatcher, Captain, Crew, Port Captain, Fleet Admin, Billing Clerk, Client (read-only).
- **Surface 2 — REST API (programmatic):** Express.js on Vercel serverless. Every UI action is an API call. Also serves AIS webhooks, Signal K pushes, QuickBooks callbacks, and port authority job intake.
- **Surface 3 — MCP layer (AI-native):** as MCP client, TugOS consumes maritime authority expertise; as MCP server, it exposes fleet data to AI agents. Inbound: USCG regs, AIS, weather, Signal K. Outbound: fleet status, crew records, job logs, fuel data.

Auth across surfaces

JWT + bcrypt with 15-minute access tokens on the web. The captain app pairs short access tokens with long-lived, device-bound refresh tokens and an offline grace policy: actions queue locally with the last-valid identity and sync on reconnect, where the refresh token re-validates before the queue commits. Offline-first (whitepaper Pillar 07) and short tokens coexist by design rather than by accident.

03 · Five MCP Servers — Adapted for Maritime Operations

- **tug-orchestration** (the brain, 20+ tools): list_vessels, get_vessel_status, list_active_jobs, get_crew_roster, trigger_dispatch, approve_fuel_alert, get_fleet_health, search_authority_catalog, get_job_briefing, generate_invoice.
- **tug-data-connectors** (the senses, 15+ tools): get_ais_position (commercial API), get_signal_k_data, get_weather_noaa, get_port_schedule, get_ais_eta, get_fuel_price, get_tidal_data, sync_quickbooks.
- **tug-notifications** (the voice, 12+ tools): send_captain_push, send_dispatch_slack, send_cert_expiry_alert, send_fuel_anomaly_alert, send_subm_audit_reminder, send_client_job_complete, send_maintenance_urgent, send_invoice_ready.
- **tug-scaffold-core** (the memory, 15+ tools): search_subchapter_m, get_tsms_requirements, search_uscg_regulations, get_stcw_requirements, get_abs_survey_standards, get_workhour_rules (46 USC 8904 / 46 CFR 15.1111), get_architecture_pattern, search_playbook.
- **tug-client-portal** (the handshake, 8 tools, client-scoped): my_jobs, my_job_history, my_invoices, my_invoice_status, my_vessels_assisted, my_compliance_certs, my_account_health, download_job_report.
- **Persistent memory (6th layer) — NotebookLM:** USCG/Sub M Inspection Playbook · Fleet Maintenance Encyclopedia · Crew Performance Records · Client Relationship History · TugOS Architecture Docs.

04 · Six AI Agents — Governed Autonomy

- **Fuel Agent** (auto-limited) — monitors per-vessel consumption vs baseline; flags anomalies >8%; approves efficiency adjustments under \$2K; higher routes to fleet manager.
- **Compliance Agent** (supervised) — daily scan of cert expiry, manning compliance, and **Sub M/TSMS record completeness** (drills due, inspections due, audit-item gaps); all recommendations require Port Captain review.
- **Dispatch Agent** (supervised) — recommends vessel assignment by AIS proximity, crew availability, bollard pull, fuel cost; dispatcher approves before push.
- **Fleet Health Agent** (autonomous) — hourly scoring on maintenance, fuel trend, compliance, utilization; Slack alert below 70.
- **Daily Briefing Agent** (autonomous, 6am) — today's jobs, crew on/off hitch, certs expiring, maintenance due, fuel vs budget, open invoices; email + Slack.
- **Invoice Agent** (autonomous) — drafts invoices from completed job logs within 1 hour; billing clerk 1-click approval.

The Interrupt Protocol. High-stakes actions hit an interrupt_before node in the LangGraph workflow; the Port Captain reviews in the War Room with full execution trace. Triggers: fuel procurement >\$5K · crew certification waivers · client billing dispute resolution · USCG deviation reporting · any crew PII change. LangGraph + MLflow governance is inherited unchanged.

05 · Seven Scheduled Automations

Task	Schedule	What it does	Connectors
daily-fleet-briefing	6am daily	Jobs, crew changes, cert expirations, fuel budget, open invoices to email + Slack	Orchestration, Notifications

Task	Schedule	What it does	Connectors
vessel-health-monitor	Hourly	Scores fleet health; Slack alert below 70; urgent push for critical maintenance	Orchestration, Notifications, Captain App
cert-expiry-sweep	7am daily	Scans STCW/MMC/drug-test/medical dates; alerts at 90/30/7-day windows	Orchestration, Notifications, Email
subm-audit-readiness	Weekly	Checks TSMS record completeness per vessel (drills, inspections, maintenance logs); flags gaps before the TPO does	Orchestration, Notifications
invoice-generation	Every 60 min	Finds completed jobs with no invoice; drafts from job log; pushes to billing queue	Orchestration, Notifications, QuickBooks
fuel-anomaly-check	Per voyage	Compares voyage fuel to baseline; flags >8% deviation	Signal K, Orchestration, Notifications
ais-position-sync	Every 5 min	Pulls live AIS for fleet vessels; updates dispatch map; ETAs for active jobs	AIS API, Data Connectors, UI WebSocket

(v1's authority-catalog-sync to Notion remains available via the template; v1's CII reporting automation is removed — IMO CII applies at 5,000 GT and above, which excludes the target fleet.)

06 · The Catalog of Authority — Maritime Authorities

Regulatory Authorities

- **46 CFR Subchapter M (Parts 136–144)** — the centerpiece: certification (COI), TSMS, operations, lifesaving, construction, and record-keeping requirements for U.S. towing vessels.
- **46 CFR Part 15** (manning; 15.1111 work hours/rest) and **46 U.S.C. 8904** (12-hour rule for licensed towing-vessel operators).
- **STCW Convention** — competency and endorsement requirements per rank.
- **ABS Rules** — survey intervals, classification standards for classed tugs.
- **IMO MARPOL** — fuel oil record-keeping where applicable by tonnage; CII explicitly out of scope for the under-5,000 GT fleet.

Data Authorities

- **Signal K** (Apache 2.0) — real-time vessel telemetry standard. **Commercial AIS APIs** (MarineTraffic/Kpler, Spire, aisstream.io — select on coverage and terms) for live positions; optional self-hosted receivers with permissively-licensed decoders. **NOAA** weather and tides. **OpenCPN** (GPLv2) chart primitives, process-isolated. **OpenShipping.org** port integration standards. The MarineTraffic AIS Toolbox (CC BY-NC-SA 4.0) may inform internal research only — it cannot ship in the product.

Financial Authorities

- **QuickBooks API** (two-way sync), **Stripe** (portal payments, ACH), **Ship & Bunker** fuel price feeds, **AWO rate benchmarks** for assist-tug market pricing.

OSL Native Integrations (reused)

- pdf-lib (MIT) — Sub M record books, work orders, invoice PDFs · PptxGenJS (MIT) — fleet reports · React Email — transactional templates · DocuSeal (AGPL-3.0) — signature capture, kept behind the adapter pattern pending a formal license review · LlamaIndex — RAG over Sub M/USCG regulations and maintenance history for the Beacon assistant.

07 · Shared Technology Stack — Inherit vs New

Layer	Status	Technology	TugOS usage
Frontend	REUSE	React 19 + Vite 5 + Tailwind	Dispatch board, crew scheduler, fuel dashboard, billing; dark maritime theme
Mobile	NEW	React Native (Expo), iOS first	Captain/crew app; offline-first with refresh-token grace; job acceptance, fuel log, photos, signatures
Backend	REUSE	Express.js 4 + Vercel serverless	OSL pattern; direct SQL with per-tenant RLS policies enforced via tenant-scoped connections
Database	CHANGE D	PostgreSQL on Supabase — dedicated TugOS project	tug_ tables: vessels, jobs, crew, fuel_logs, maintenance, invoices, certs, clients (+ tug_agent_runs); isolation for SOC 2 and port-authority reviews
Auth	REUSE+	JWT + bcrypt, 15-min web tokens + mobile refresh grace	Roles: fleet_admin, port_captain, dispatcher, captain, crew, billing, client
AI Service	REUSE	callAI + Paperclip	Dual-model setup powers the TugOS Beacon and the six maritime agents
IoT Bridge	NEW	Signal K Server (Apache 2.0)	Fuel flow meters, GPS, engine monitors to TugOS via WebSocket
AIS Tracking	NEW	Commercial AIS API (+ optional receivers)	Live dispatch map overlay, ETAs; license-clean by construction
Agent Framework	REUSE	LangGraph + MLflow	Interrupt Protocol; all runs logged to tug_agent_runs
Deployment	REUSE	Vercel + Supabase + GitHub CI/CD	Auto-deploy on push; Fly.io for compute-heavy Signal K bridge workers if needed

08 · The OSL Network Effect — How TugOS Compounds the Moat

Each new OSL vertical enriches the Catalog of Authority. TugOS launches as App 24 and adds the maritime authorities — Subchapter M, TSMS, STCW, ABS, Signal K — growing the catalog from 15 to 25+ entries and making every future marine vertical (PortOS, MarineInspector) cheaper to build. The flywheel compounds: each vertical means less time to the next.

09 · Build vs Reuse — The Honest Accounting

Inherited from the OSL genetic template

StandardHeader and Scaffold Beacon · role-based sidebar + requireRole middleware · JWT auth + session management · Supabase connection layer + RLS patterns · all @openscaffold/integrations adapters · Vercel pipeline + GitHub CI/CD · LangGraph + MLflow governance · Paperclip orchestration · MCP server SDK + five-server pattern · Interrupt Protocol · callAI utility · theme system · CRUD standards.

Net new for TugOS

Dedicated Supabase project + tug_ schema (9 tables) · Signal K IoT bridge · commercial AIS integration + dispatch map · React Native captain app with offline-first sync · maritime Catalog of Authority entries (Sub M, TSMS, 46 CFR 15, STCW, ABS) · dispatch board UI · hitch scheduler · Sub M/TSMS compliance tracker · per-voyage fuel log + anomaly logic · tug billing engine (hourly / bollard pull / charter) · six maritime AI agents · QuickBooks two-way sync · NOAA weather + tidal connector · mobile refresh-token auth design.

